For Examiner's Use

5 **(a)** Fig. 5.1 shows a gold-leaf electroscope. The gold leaf is deflected because the electroscope is negatively-charged.

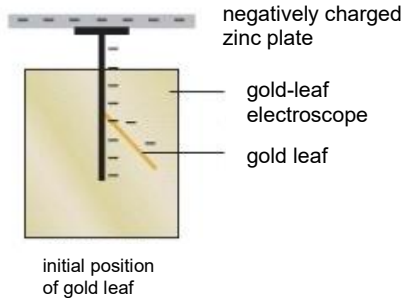

Fig. 5.1

(i) When red light is incident on the electroscope, the gold leaf remains deflected as shown in Fig. 5.2 a.

When ultra-violet radiation is incident on the electroscope, the gold leaf falls as shown in Fig. 5.2 b. This is due to a decrease in the negative charge of the gold leaf.

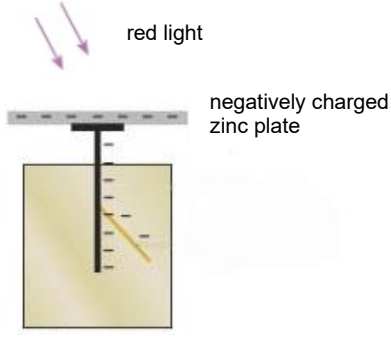

Fig. 5.2 a

Fig. 5.2 b

Using Fig. 5.2 a and Fig. 5.2 b, explain how this is a demonstration of the photoelectric effect.

……

……

……

……

…… [2]

For Examiner's Use

(ii) When the intensity of the red light that is incident on the electroscope in Fig. 5.2 a is increased, the deflection of the gold leaf remains the same.

Explain how this is an evidence for the particulate nature of electromagnetic radiation.

……

……

……

……

…… [2]

(b) X-rays are produced by bombarding electrons, accelerated through a high potential difference, on a metal target. A typical X-ray spectrum is shown in Fig. 5.3.

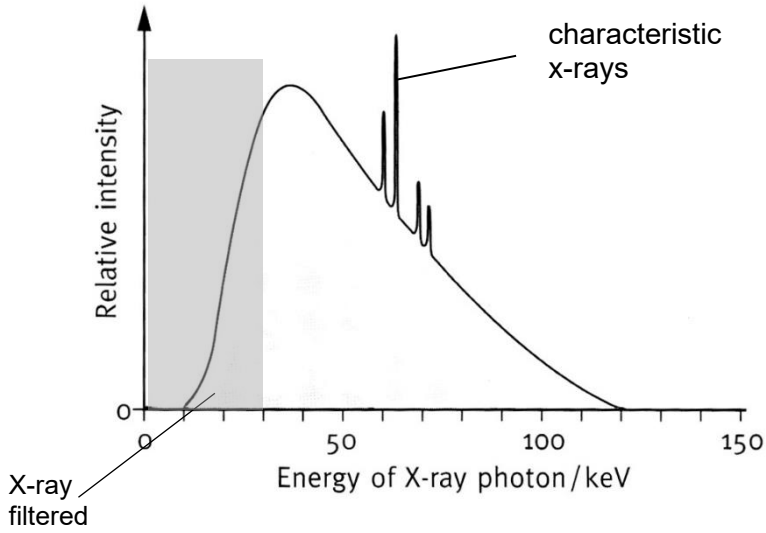

Fig. 5.3

(i) State the potential difference used to accelerate the electrons to produce the X-rays.

potential difference = …… V [1]

For Examiner's Use

(ii) Explain the mechanism by which the characteristic X-rays are produced.

……

……

……

……

…… [2]

(iii) Explain why characteristic X-rays produced are unique to the element used in the metal target.

……

……

…… [1]

(iv) X-rays can be used as a diagnostic tool. This is possible because different parts of our body absorb different amounts of X-ray. The lower energy X-rays are often filtered as shown in Fig. 5.3.

Suggest a reason why this is necessary.

……

……

…… [1]

For Examiner's Use

(c) An electron travelling at 1200 m s^{-1} passes through a slit of 24 μm as shown in Fig. 5.4.

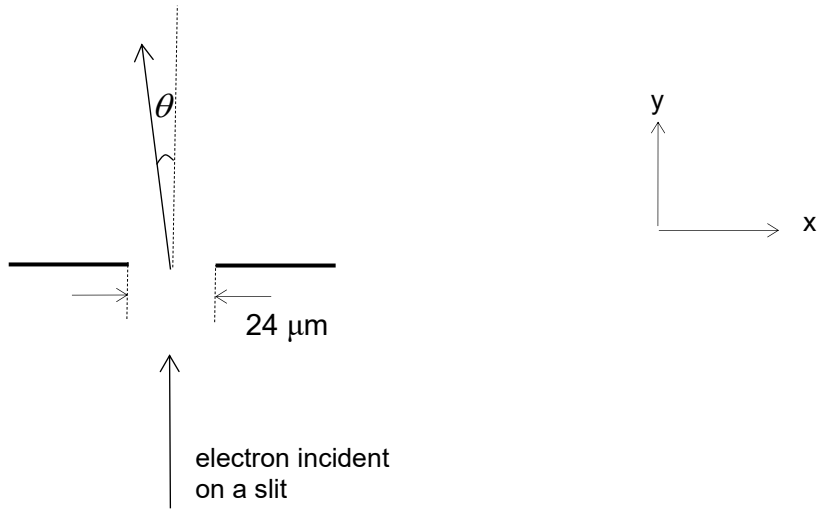

Fig. 5.4

(i) Calculate the minimum uncertainty of the momentum of the electron in the x-direction, Δp_x.

Δp_x = ... kg m s^{-1} [2]

(ii) In an experiment where many electrons are put through the slit, it is observed that most electrons deviate from its original path within the angle θ as shown in Fig. 5.4 due to the uncertainty of the electrons' momentum.

Use your answer in **(c)(i)** or otherwise, calculate θ.

θ = ... ° [3]